

21/04

3. Probability Distribution

- Binomial Distribution
- Poisson Distribution
- Normal Distribution

Binomial Distribution

A Random variable X as Binomial distribution if it assumes only non negative values and its probability distribution is given by

$$P(X=x) = \begin{cases} nC_x p^x q^{n-x}, & x=0,1,2,\dots,n \\ 0 & , \text{ elsewhere} \end{cases}$$

Where, n = number of trials

p = number of success

q = number of failures

$$p+q=1$$

*** Find the moment generating function for the binomial distribution and also find mean and variance

We know that,

$$M_X(t) = E[e^{tx}]$$

$$= \sum_{x=0}^n e^{tx} p(x=x)$$

$$= \sum_{x=0}^n e^{tx} n C_x p^x q^{n-x}$$

$$= \sum_{x=0}^n n C_x (pe^t)^x q^{n-x}$$

$$M_x(t) = (q + pe^t)^n \quad \left[\because (x+a)^n = \text{Binomial Series} \right]$$

To find mean (or) μ' (or) $\bar{x}$

We know that

$$\mu' = \frac{d}{dt} [M_x(t)] \text{ when } t=0$$

$$= \frac{d}{dt} [(q + pe^t)^n] \therefore t=0$$

$$= [n(q + pe^t)^{n-1} (0 + pe^t)] \therefore t=0$$

$$= [npe^t (q + pe^t)^{n-1}] \therefore t=0$$

$$= npe^0 (q + pe^0)^{n-1}$$

$$= np(q + p)^{n-1}$$

$$= np(1)^{n-1}$$

$$= np$$

$$\boxed{\begin{aligned} \mu' &= np \\ \text{mean} = \bar{x} &= np \end{aligned}}$$

To find the variance of x

$$\text{i.e., } \text{Var}(x) = \mu_2' - (\mu_1')^2$$

$$\text{Var}(x) = E(x^2) - (E(x))^2$$

To find μ_2' :

We know that

$$\mu_2' = \frac{d^2}{dt^2} [M_x(t)] \quad \therefore t=0$$

$$= \frac{d}{dt} \left[\frac{d}{dt} (M_x(t)) \right] \quad \therefore t=0$$

$$= \frac{d}{dt} [np e^t (q + p e^t)^{n-1}] \quad \therefore t=0$$

$$= np \frac{d}{dt} [e^t (q + p e^t)^{n-1}] \quad \therefore t=0$$

$$[\because d(uv) = u dv + v du]$$

$$= np [e^t (q + p e^t)^{n-1} + e^t (n-1) (q + p e^t)^{n-2} (0 + p e^t)] \quad \therefore t=0$$

$$= np [e^0 (q + p e^0)^{n-1} + e^0 (n-1) (q + p e^0)^{n-2} (0 + p e^0)]$$

$$= np [1^{n-1} + 1(n-1)(1)^{n-2}(p)]$$

$$= np(1 + (n-1)p)$$

$$= np[1 + np - p]$$

$$= np + n^2 p^2 - np^2$$

$$\text{Var}(x) = \mu_2' - \mu_1'^2$$

$$= np + n^2 p^2 - np^2 - (np)^2$$

$$= np + n^2 p^2 - np^2 - n^2 p^2$$

$$= np - np^2$$

$$= np(1-p)$$

$$\boxed{\text{var}(x) = npq}$$

$$[1-p = q]$$

Note :

$$\text{let } x \sim B(n, p)$$

$$\text{mean} = np$$

$$\text{Variance} = npq$$

1. A coin is tossed 9 times Find the probability of getting 5 heads

$$S = \{H, T\}$$

$$\text{let } x \sim B(n, p)$$

$$n = 9$$

$$p = 1/2 \quad P(\text{getting head})$$

$$q = 1 - p = 1 - 1/2 = 1/2$$

$$P(x = 5) = ?$$

$$P(x = x) = {}^n C_x p^x q^{n-x}$$

$$P(x = 5) = {}^9 C_5 (1/2)^5 (1/2)^{9-5}$$

$$= {}^9 C_5 (1/2)^5 (1/2)^4$$

$$= {}^9 C_5 (1/2)^{5+4}$$

$$= {}^9 C_5 (1/2)^9 = \frac{66}{256} \approx 0.2460$$

2. 10 coins are thrown simultaneously and the probability of getting at least 7 heads

$$S = \{H, T\}$$

$$\text{Let } X \sim B(n, p)$$

$$n = 10$$

$$p = 1/2, q = 1/2$$

$$P(X \geq 7) = ?$$

$$\left\{ \begin{array}{l} P(\text{at least } a) = P(X \geq a) \\ P(\text{at most } a) = P(X \leq a) \\ P(\text{exact } a) = P(X = a) \end{array} \right\}$$

$$P(X = x) = {}^n C_x p^x q^{n-x}$$

$$P(X \geq 7) = P(X = 7, 8, 9, 10)$$

$$= P(X = 7) + P(X = 8) + P(X = 9) + P(X = 10)$$

$$= {}^{10}C_7 \left(\frac{1}{2}\right)^7 \left(\frac{1}{2}\right)^{10-7} + {}^{10}C_8 \left(\frac{1}{2}\right)^8 \left(\frac{1}{2}\right)^{10-8} +$$

$${}^{10}C_9 \left(\frac{1}{2}\right)^9 \left(\frac{1}{2}\right)^{10-9} + {}^{10}C_{10} \left(\frac{1}{2}\right)^{10} \left(\frac{1}{2}\right)^{10-10}$$

$$= \frac{176}{4} = \underline{\underline{0.1718}}$$

$$\downarrow$$

$$0.000176$$

3. The probability that the life of a Bulb is 100 days is 0.05. Find the probability that out of 6 bulbs

- (i) at least one (ii) none (iii) greater than 4
- (iv) between 1 and 3

Let $X \sim B(n, p)$

We know that

$$P(X=x) = {}^n C_x p^x q^{n-x}, \quad x=0, 1, 2, \dots, n$$

Here $n=6$ (bulbs)

$$p=0.05, \quad q=1-p=0.95 = 19/20$$

$$(i) \quad P(X \geq 1)$$

$$P(X \geq a) = 1 - P(X < a)$$

$$P(X \geq 1) = 1 - P(X < 1)$$

$$= 1 - P(X=0)$$

$$= 1 - {}^6 C_0 \left(\frac{1}{20}\right)^0 \left(\frac{19}{20}\right)^{6-0}$$

$$= 1 - \left(\frac{19}{20}\right)^6$$

$$= 1 - 0.73509$$

$$P(X \geq 1) = \underline{0.2649}$$

$$(ii) \quad P(X=0) = {}^6 C_0 \left(\frac{1}{20}\right)^0 \left(\frac{19}{20}\right)^{6-0}$$

$$P(X=0) = (1)(1) \left(\frac{19}{20}\right)^6$$

$$P(X=0) = \underline{0.73509}$$

$$(iii) \quad P(X \leq 4) = P(X=5, 6)$$

$$= P(X=5) + P(X=6)$$

$$= {}^6 C_5 \left(\frac{1}{20}\right)^5 \left(\frac{19}{20}\right)^{6-5} + {}^6 C_6 \left(\frac{1}{20}\right)^6 \left(\frac{19}{20}\right)^{6-6}$$

$$= \frac{115}{(20)^6} = 0.$$

$$\begin{aligned} \text{(iv)} \quad P(1 \leq x \leq 3) &= P(x=1, 2, 3) \\ &= P(x=1) + P(x=2) + P(x=3) \\ &= {}^6C_1 \left(\frac{1}{20}\right)^1 \left(\frac{19}{20}\right)^{6-1} + {}^6C_2 \left(\frac{1}{20}\right)^2 \left(\frac{19}{20}\right)^{6-2} \\ &\quad + {}^6C_3 \left(\frac{1}{20}\right)^3 \left(\frac{19}{20}\right)^{6-3} \\ &= \frac{(19)^3}{(20)^6} \quad (24 \neq 1) = 0. \end{aligned}$$

4. 20% of items produced from a factory are defective. find the probability that in a sample of 5 chosen at random

(i) none is defective $P(x=0)$

(ii) one is defective $P(x=1)$

(iii) probability of $P(1 < x < 4)$. $P(x=2, 3)$

Sol $n = 5$

$$p = \frac{20}{100} = \frac{1}{5}, \quad q = 1 - p = 1 - \frac{1}{5} = \frac{4}{5}$$

$$\begin{aligned} \text{(i)} \quad P(x=0) &= {}^5C_0 (0.2)^0 (0.8)^{5-0} \\ &= (0.8)^5 = 0.32768 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad P(x=1) &= {}^5C_1 (0.2)^1 (0.8)^{5-1} \\ &= 5(0.2)(0.8)^4 \\ &= 5 \times 0.2 \times 0.4096 \\ &= 5 \times 0.08192 \\ &= 0.4096 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad p(x=2,3) &= p(x=2) + p(x=3) \\
 &= {}^5C_2 (0.2)^2 (0.8)^{5-2} + {}^5C_3 (0.2)^3 (0.8)^{5-3} \\
 &= \frac{5!}{2!3!} \times 0.04 \times 0.512 + \frac{5!}{3!2!} \times 0.008 \times 0.008 \\
 &= \frac{5 \times 4 \times 2}{2} \times 0.02048 + \frac{5 \times 4 \times 2}{2} \times 0.00512 \\
 &= 0.2048 + 0.0512 = 0.2558
 \end{aligned}$$

5) The mean and variance of B.D are 4 and $4/3$ respectively then find $p(x=0)$ and $p(x \geq 2)$

sol let $x \sim B(n, p)$

We know that mean $= np$

Variance $= npq$

We know that

$$p(x=x) = {}^nC_x p^x q^{n-x} \quad x=0,1,2,3, \dots$$

$$\text{mean} = 4$$

$$np = 4 \rightarrow \textcircled{1}$$

$$\text{Variance} = 4/3$$

$$npq = 4/3 \rightarrow \textcircled{2}$$

Substituting $\textcircled{1}$ in $\textcircled{2}$

$$\frac{\textcircled{1}}{\textcircled{2}} \quad \frac{np}{npq} = \frac{4}{4/3}$$

$$\frac{1}{q} = \frac{12}{4}$$

$$\frac{1}{q} = 3$$

$$\boxed{\frac{1}{3} = q}$$

then

$$p + q = 1$$

$$p + \frac{1}{3} = 1$$

$$p = 1 - \frac{1}{3}$$

$$\boxed{p = \frac{2}{3}}$$

$$np = 4$$

$$n\left(\frac{2}{3}\right) = 4$$

$$\frac{2n}{3} = 4$$

$$2n = 12$$

$$\boxed{n = 6}$$

$$\therefore n = 6, p = \frac{2}{3}, q = \frac{1}{3}$$

We know that

$$P(X \geq 1) = 1 - P(X < 1)$$

$$= 1 - P(X = 0)$$

$$= 1 - {}^6C_0 \left(\frac{2}{3}\right)^0 \left(\frac{1}{3}\right)^{6-0}$$

$$= 1 - \left(\frac{1}{3}\right)^6$$

$$= 1 - \frac{1}{3^6}$$

$$= 1 - \frac{1}{729} = \frac{728}{729} = 0.9986$$

$$P(X \geq 2) = 1 - P(X < 2)$$

$$= 1 - P(X = 0, 1)$$

$$= 1 - P(X = 0) - P(X = 1)$$

$$= 1 - \left(\frac{1}{3}\right)^6 + {}^6C_1 \left(\frac{2}{3}\right)^1 \left(\frac{1}{3}\right)^{6-1}$$

$$= 1 - \frac{1}{3^6} + 6 \times \frac{2}{3} \times \frac{1}{3^5}$$

$$= 1 - \frac{728}{729}$$

$$= 1 - \frac{1}{729} + \frac{4}{243} = 0.98213$$

$$= \cancel{0.9986} + 0.0165 =$$

6) The mean of B.D is 3 and the variance is $9/4$ Find

(i) The value of n ($n=12$, $p=1/4$, $q=3/4$)

(ii) $P(X \geq 7)$ [0.1446]

(iii) $P(1 \leq X \leq 6)$ [0.82]

So let $X \sim B(n, p)$

$$\text{mean} = np$$

$$\text{Variance} = npq$$

We know that

$$P(X=x) = {}^nC_x p^x q^{n-x}$$

$$\text{mean} = 3, \text{Variance} = 9/4$$

$$np = 3 \quad \text{--- (1)}$$

$$npq = 9/4 \quad \text{--- (2)}$$

Substituting (1) in (2)

$$\frac{(1)}{(2)} \frac{np}{npq} = \frac{3}{9/4}$$

$$\frac{1}{q} = \frac{12/4}{9/3}$$

$$\frac{1}{q} = 4/3$$

$$\frac{1}{q} = \frac{4}{3}$$

$$3 = 4q$$

$$\boxed{q = 3/4}$$

$$p + q = 1$$

$$p + \frac{3}{4} = 1$$

$$p = 1 - \frac{3}{4}$$

$$p = \frac{4-3}{4}$$

$$\boxed{p = 1/4}$$

$$(1) \Rightarrow np = 3$$

$$n\left(\frac{1}{4}\right) = 3$$

$$\frac{n}{4} = 3 \Rightarrow \boxed{n = 12}$$

$$p(x \geq 7) = 1 - p(x < 7)$$

$$= 1 - p(x = 0, 1, 2, 3, 4, 5, 6)$$

$$= 1 - 12C_0 \left(\frac{1}{4}\right)^0 \left(\frac{3}{4}\right)^{12-0} + 12C_1 \left(\frac{1}{4}\right)^1 \left(\frac{3}{4}\right)^{11} +$$

$$12C_2 \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^{10} + 12C_3 \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^9 +$$

$$12C_4 \left(\frac{1}{4}\right)^4 \left(\frac{3}{4}\right)^8 + 12C_5 \left(\frac{1}{4}\right)^5 \left(\frac{3}{4}\right)^7 +$$

$$12C_6 \left(\frac{1}{4}\right)^6 \left(\frac{3}{4}\right)^6$$

$$= 0.1446$$

$$p(1 \leq x \leq 6) = p(x = 2, 3, 4, 5)$$

$$= 12C_2 \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^{10} + 12C_3 \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^9$$

$$+ 12C_4 \left(\frac{1}{4}\right)^4 \left(\frac{3}{4}\right)^8 + 12C_5 \left(\frac{1}{4}\right)^5 \left(\frac{3}{4}\right)^7$$

$$= \underline{\underline{0.82}}$$

Binomial Expected Frequency distribution

Let us suppose that n trials constitute an experiment then if this experiment is repeated N times the frequency function after binomial distribution is given by

$$f(x) = N p(x=x)$$

$$= N \cdot nC_x p^x q^{n-x}$$

$$x = 0, 1, 2, 3, \dots, n$$

1) Find a binomial distribution to the following data

x	0	1	2	3	4	5
f	2	14	20	34	22	8

Sol Let $X \sim B(n, p)$

$$P(X=x) = {}^N n C_x p^x q^{n-x} \quad \begin{matrix} [x=0, 1, 2, 3, 4, 5 \dots] \\ [x=0, 1, 2, 3, \dots, n] \end{matrix}$$

Here $n=5$

We know that

$$f(x) = {}^N n C_x p^x q^{n-x}$$

$$N = \sum f_i = 2 + 14 + 20 + 34 + 22 + 8 = 100$$

$$\begin{aligned} \bar{x} &= \frac{\sum f_i x_i}{\sum f_i} = \frac{0(2) + 1(14) + 2(20) + 3(34) + 4(22) + 5(8)}{100} \\ &= \frac{14 + 40 + 102 + 88 + 40}{100} = 2.84 \end{aligned}$$

$$\bar{x} = 2.84$$

$$\text{mean} = 2.84$$

$$np = 2.84$$

$$5p = 2.84$$

$$p = \frac{2.84}{5}$$

$$\boxed{p = 0.568}$$

$$q = 1 - p$$

$$q = 1 - 0.568$$

$$\boxed{q = 0.432}$$

$$f(0) = 100 \times {}^5 C_0 (0.568)^0 (0.432)^{5-0}$$

$$= 100 \times (0.432)^5 = 100(0.0150)$$

$$= 1.5 \approx 2$$

$$\begin{aligned}
 f(1) = n=1 &= 100 \times {}^5C_1 (0.568)^1 (0.432)^{5-1} \\
 &= 100 \times 5 (0.568) (0.03482) \\
 &= 9.88 \approx 10
 \end{aligned}$$

$$\begin{aligned}
 n=2 &= 100 \times {}^5C_2 \times (0.568)^2 \times (0.432)^{5-2} \\
 &= 100 \times \frac{5!}{2!3!} \times (0.322) \times (0.08) \\
 &= 25.76 \approx 26
 \end{aligned}$$

$$\begin{aligned}
 n=3 &= 100 \times {}^5C_3 \cdot (0.568)^3 \times (0.432)^2 \\
 &= 100 \times \frac{5!}{3!2!} (0.1832) (0.186) \\
 &= 100 \times \frac{5 \times 4 \times 3}{2} \times 0.0348 \\
 &= 100 \times 10 \times 0.034 \\
 &= 34 \approx 34
 \end{aligned}$$

$$\begin{aligned}
 n=4 &= 100 \times {}^5C_4 \times (0.568)^4 \times (0.432)^1 \\
 &= 100 \times \frac{5!}{4!1!} \times 0.104 \times 0.432 \\
 &= 22.465 \approx 22
 \end{aligned}$$

$$\begin{aligned}
 n=5 &= 100 \times {}^5C_5 \times (0.568)^5 \times (0.432)^0 \\
 &= 100 \times 1 (0.0568)^5 \\
 &= 5.91 \approx 6
 \end{aligned}$$

We know that the expected frequency Binomial distribution

$$f(x) = N p(x=x) = N n C_x p^x q^{n-x}$$

2) Fit a binomial distribution to the following data

x	0	1	2	3	4	5	6
f	13	25	52	58	32	16	4

Sol let $x \sim B(n, p)$ $x = 0, 1, 2, 3, 4, 5, 6$

$$P(X=x) = {}^N n C_x p^x q^{n-x} \quad x = 0, 1, 2, 3, \dots, n$$

Here $n = 6$

We know that

$$f(x) = {}^N n C_x p^x q^{n-x}$$

$$N = \sum f_i = 13 + 25 + 52 + 58 + 32 + 16 + 4 = 200$$

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{535}{200} = 2.675$$

$$\text{mean} = 2.675$$

$$np = 2.675$$

$$6p = 2.675$$

$$p = \frac{2.675}{6} = 0.445$$

$$q = 1 - p$$

$$q = 1 - 0.445$$

$$q = 0.555$$

$$n = 6, p = 0.445, q = 0.555$$

We know that the expected Frequency
Binomial distribution $f(x) = N P(X=x)$

$$= N {}^n n C_x p^x q^{n-x}$$

$$n=0 \Rightarrow 5.84$$

$$n=1 \Rightarrow f(1) = 28.0884$$

Sol

$$x=2 \Rightarrow f(2) = 56.3112$$

$$x=3 \Rightarrow f(3) = 59.84$$

$$x=4 \Rightarrow f(4) = 36.2208$$

$$x=5 \Rightarrow f(5) = 11.5884$$

$$x=6 \Rightarrow f(6) = 1.552$$

Poisson Distribution : A Random variable X is said to follow poisson distribution if it assumes only non-negative values and its probability distribution is given by

$$p(X=x) = \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!} & x=0, 1, 2, \dots, \infty \\ 0 & \text{otherwise} \end{cases}$$

★ Q Find the moment generating function for the poisson distribution also find mean and variance

Sol: We know that

$$M_X(t) = \sum_{x=0}^{\infty} e^{tx} p(X=x)$$

$$= \sum_{x=0}^{\infty} e^{tx} \frac{e^{-\lambda} \lambda^x}{x!}$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} \frac{e^{tx} \lambda^x}{x!}$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} \frac{(\lambda e^t)^x}{x!}$$

$$= e^{-\lambda} \left[\frac{(\lambda e^t)^0}{0!} + \frac{(\lambda e^t)^1}{1!} + \frac{(\lambda e^t)^2}{2!} + \dots \right]$$

$$= e^{-\lambda} \left[\frac{1}{1} + \frac{\lambda e^t}{1!} + \frac{(\lambda e^t)^2}{2!} + \frac{(\lambda e^t)^3}{3!} + \dots \right]$$

$$= e^{-\lambda} \left[1 + \frac{\lambda}{1} + \frac{\lambda^2}{2!} + \frac{\lambda^3}{3!} + \dots \right]$$

$$= e^{-\lambda} e^{\lambda} \left[\because e^{\lambda} = 1 + \frac{\lambda}{1} + \frac{\lambda^2}{2!} + \dots \right]$$

$$= e^{-\lambda} e^{\lambda}$$

$$= e^{\lambda - \lambda}$$

$$[\lambda e^t = \lambda]$$

$$= e^{-\lambda + \lambda}$$

$$= e^{-\lambda + \lambda e^t}$$

$$= e^{\lambda(e^t - 1)}$$

$$= e^{-\lambda(1 - e^t)}$$

$$M_X(t) = e^{-\lambda(1 - e^t)}$$

To find the mean (or) μ (or) $E(X)$ μ

We know that

$$\mu_1' = \frac{d}{dt} [M_X(t)] \bigg|_{t=0}$$

$$= \frac{d}{dt} [e^{\lambda(e^t - 1)}] \bigg|_{t=0}$$

$$= [e^{\lambda(e^t - 1)} \cdot \lambda(e^t - 1)] \bigg|_{t=0}$$

$$= [\lambda e^t e^{\lambda(e^t - 1)}] \bigg|_{t=0}$$

$$= \lambda e^0 e^{\lambda(e^0 - 1)}$$

$$= \lambda \cdot 1 \cdot e^{\lambda(1 - 1)}$$

$$= \lambda e^{\lambda(0)}$$

$$= \lambda e^{\lambda(0)}$$

$$= \lambda e^0 = \lambda(1) = \lambda \Rightarrow \boxed{\mu_1' = \lambda \text{ mean}}$$

To find Variance of X

We know that

$$\text{Var}(X) = \mu_2' - (\mu_1')^2$$

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

To find μ_2'

We know that

$$\mu_2' = \frac{d^2}{dt^2} [M_X(t)] \big|_{t=0}$$

$$= \frac{d}{dt} \left[\frac{d}{dt} (M_X(t)) \right] \big|_{t=0}$$

$$= \frac{d}{dt} [\lambda e^t \lambda (e^t - 1)] \big|_{t=0}$$

$$= \lambda \frac{d}{dt} [e^t e^{\lambda(e^t - 1)}] \big|_{t=0}$$

$\downarrow \quad \downarrow$
 $u \quad v$

$$d(u, v) = u dv + v du$$

$$= \lambda [e^t e^{\lambda(e^t - 1)} \cdot \lambda (e^t - 0) + e^{\lambda(e^t - 1)} e^t]$$

$$= \lambda [e^0 e^{\lambda(e^0 - 1)} \cdot \lambda (e^0 - 0) + e^{\lambda(e^0 - 1)} e^0]$$

$$= \lambda [1 \cdot 1 \cdot \lambda (1 - 0) + e^0 \cdot e^0]$$

$$= \lambda (\lambda + 1)$$

$$= \lambda^2 + \lambda$$

$$\mu_2' = \lambda^2 + \lambda$$

$$\text{Var } X = \mu_2' - (\mu_1')^2$$

$$= \lambda^2 + \lambda - (\lambda)^2 = \lambda$$

$$S.D = \sqrt{\text{Var } X}$$

$$S.D = \sqrt{\lambda}$$

Q 2% of the items of a factory are defective. The items are packed in boxes what is the probability that there will be

(i) 2 defective items

(ii) atleast 3 defective items in a box of 100 items

Sol Given $p = 2\% = \frac{2}{100} = 0.02$, $n = 100$

We know that

$\lambda = np$ [limiting case of poisson distribution]

$$\lambda = 100 \times 0.02$$

$$\lambda = 2$$

Let $x \sim p(\lambda)$

We know that

$$p(x=x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad x = 0, 1, 2, \dots, \infty$$

$$(i) \quad p(x=2) = \frac{e^{-2} (2)^2}{2!} = \frac{4}{2} e^{-2} = 2e^{-2} = \frac{2}{e^2}$$

$$(ii) \quad p(x \geq 3) = 1 - p(x < 3)$$

$$= 1 - p(x=0) + p(x=1) + p(x=2)$$

$$= 1 - \left[\frac{e^{-2} (2)^0}{0!} + \frac{e^{-2} (2)^1}{1!} + \frac{e^{-2} (2)^2}{2!} \right]$$

$$= 1 - e^{-2} \left(1 + 2 + \frac{2}{2} \right)$$

$$= 1 - e^{-2} (1 + 2 + 2)$$

$$= 1 - e^{-2} (5) = 1 - 5e^{-2} = 0.3233 //$$

3) If a poisson distribution in such that
 $p(x=1) = p(x=3)$ then find (i) $p(x \geq 1)$, mean
 (ii) $p(x \leq 3)$ (iii) $p(2 \leq x \leq 5)$

sol let $x \sim p(n, p)$

We know that

$$p(x=x) = \frac{e^{-\lambda} \lambda^x}{x!}, x=0, 1, 2, \dots, \infty$$

Given

$$\frac{3}{2} p(x=1) = p(x=3)$$

$$\frac{3}{2} \frac{e^{-\lambda} \lambda^1}{1!} = \frac{e^{-\lambda} \lambda^3}{3!}$$

$$\frac{3}{2} \frac{\lambda}{1} = \frac{\lambda^3}{6} \quad (3 = 3 \times 2 \times 1)$$

$$\frac{3\lambda}{2} = \frac{\lambda^3}{6}$$

$$\frac{3}{2} = \frac{\lambda^2}{6}$$

$$\frac{3}{1} = \frac{\lambda^2}{3}$$

$$9 = \lambda^2$$

$$\lambda = \pm 3$$

$$\boxed{\text{mean} = \lambda = 3}$$

$$(i) p(x \geq 1) = 1 - p(x < 1)$$

$$= 1 - p(x=0)$$

$$= 1 - \frac{e^{-3}(3)^0}{0!}$$

$$= 1 - e^{-3} = 0.9502$$

3) If X is a poisson Variate such that
 $p(X=2) = 9p(X=4) + 90p(X=6)$. Find the mean
 and Variance.

Sol Given

$$p(X=2) = 9p(X=4) + 90p(X=6)$$

We know that let $X \sim p(\lambda)$

$$p(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}, x=0,1,2,3, \dots, \infty$$

$$\frac{e^{-\lambda} \lambda^2}{2!} = 9 \frac{e^{-\lambda} \lambda^4}{4!} + 90 \frac{e^{-\lambda} \lambda^6}{6!}$$

$$\frac{\lambda^2}{2} = 9 \frac{\lambda^4}{24} + 90 \frac{\lambda^6}{720}$$

$$\frac{\lambda^2}{2} = \lambda^2 \left[\frac{9}{24} \lambda^2 + \frac{90}{720} \lambda^4 \right]$$

$$\frac{1}{2} = \frac{9}{24} \lambda^2 + \frac{90}{720} \lambda^4$$

$$\frac{1}{2} = \frac{3}{8} \lambda^2 + \frac{1}{8} \lambda^4$$

$$\frac{1}{2} = \frac{3\lambda^2 + \lambda^4}{8}$$

$$1 = \frac{3\lambda^2 + \lambda^4}{4}$$

$$4 = 3\lambda^2 + \lambda^4$$

$$\lambda^4 + 3\lambda^2 - 4 = 0$$

$$\lambda^4 + 3\lambda^2 - 4 = 0$$

$$\cancel{\lambda^4 + 4\lambda^2}$$

$$\text{let } \lambda^2 = t$$

$$(t^2)^2 + 3t^2 - 4 = 0$$

$$t^2 + 3t - 4 = 0$$

$$t^2 + 3t - 4 = 0$$

$$t^2 + 4t - t - 4 = 0$$

$$t(t+4) - 1(t+4) = 0$$

$$t(t+4)(t-1) = 0$$

$$(t+4)(t-1) = 0$$

$$t+4=0, t-1=0$$

$$t = -4, 1$$

$$t = -2i, \quad t = 1$$

$$\lambda^2 = -4, \quad \lambda^2 = 1$$

$$\lambda^2 = -4, \quad \lambda = \pm 2i$$

$$\lambda^2 = 4i^2, \quad \boxed{\lambda = 1}$$

$$\lambda = \pm 2i,$$

4) Fit a poisson distribution to the following data and calculate the expected Frequency

$$x = 0 \quad 1 \quad 2 \quad 3 \quad 4$$

$$f: 46 \quad 38 \quad 22 \quad 9 \quad 1$$

sol We know that,

The expected frequency for the poisson distribution

$$= N p(x=x)$$

$$= N \frac{e^{-\lambda} \lambda^x}{x!}$$

$$\text{Where } N = \sum f_i$$

$$\text{To find } N = 46 + 38 + 22 + 9 + 1 = 116$$

To find λ (or) $\bar{x}$ (or) mean

Given data is Grouped

WKT

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{(0 \times 46) + (1 \times 38) + (2 \times 22) + (3 \times 9) + (4 \times 1)}{116}$$

$$\bar{x} = \frac{113}{116} = 0.974$$

$$\boxed{\lambda = 0.974}$$

To find expected Frequency = $N(P(x=x))$

$$f(x) = N \frac{e^{-\lambda} \lambda^x}{x!}$$

$$f(x) = \frac{116 e^{-0.974} (0.974)^x}{x!}$$

$$x=0, f(0) = \frac{116 e^{-0.974} (0.974)^0}{0!} = 44$$

$$x=1, f(1) = \frac{116 e^{-0.974} (0.974)^1}{1!} = 42$$

$$x=2, f(2) = \frac{116 e^{-0.974} (0.974)^2}{2!} = 20.7$$

$$x=3, f(3) = \frac{116 e^{-0.974} (0.974)^3}{3!} = 7$$

$$x=4, f(4) = \frac{116 e^{-0.974} (0.974)^4}{4!} = \underline{\underline{2}}$$

x	0	1	2	3	4
f	44	38	22	9	1
f(x)	44	42	21	7	2

5. Fit a poisson distribution

x	0	1	2	3	4	5	6	7	8	9	10
f	1	4	15	22	21	20	8	6	2	0	1

Recurrence relation for Poisson Distribution

Let $x \sim P(\lambda)$

We know that

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$\text{Now } P(X=x+1) = \frac{e^{-\lambda} \lambda^{x+1}}{(x+1)!}$$

$$= \frac{e^{-\lambda} \lambda^x \lambda}{(x+1)x!}$$

$$= \left(\frac{\lambda}{x+1}\right) \frac{e^{-\lambda} \lambda^x}{x!}$$

$$P(X=x+1) = \left(\frac{\lambda}{x+1}\right) \cdot P(X=x)$$

$$\therefore P(X=x+1) = \left(\frac{\lambda}{x+1}\right) P(X=x)$$

1. Use recurrence relation to find probabilities $x=0, 1, 2, 3, 4, 5$ with mean 3.

Sol. Let $x \sim P(\lambda)$

We know that, $P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$ $x=0, 1, 2, 3, 4, 5$

Given mean $= \lambda = 3$

To find:

$x=x$	0	1	2	3	4	5
$P(X=x)$						

$$P(X=0) = \frac{e^{-\lambda} \lambda^x}{x!} = \frac{e^{-3} 3^0}{0!} = e^{-3} = 0.0498$$

WKT

$$P(X=x+1) = \left(\frac{\lambda}{x+1} \right) P(X=x)$$

put $x=0$,

$$\begin{aligned} P(X=1) &= \left(\frac{\lambda}{0+1} \right) P(X=0) \\ &= \frac{3}{1} \times 0.0498 = 0.1494 \end{aligned}$$

put $x=1$

$$\begin{aligned} P(X=2) &= \left(\frac{\lambda}{1+1} \right) P(X=1) \\ &= \left(\frac{3}{2} \right) (0.1494) = 0.2241 \end{aligned}$$

put $x=2$

$$P(X=3) = \left(\frac{3}{3} \right) (0.2241) = 0.2241$$

put $x=3$

$$P(X=4) = \frac{3}{4} (0.2241) = 0.1681$$

put $x=4$

$$P(X=5) = \frac{3}{5} (0.1681) = 0.1009$$

put $x=5$

$$P(X=6) = \frac{3}{6} ($$

Therefore, the probabilities are

x	0	1	2	3	4	5
$P(X=x)$	0.0498	0.1494	0.2241	0.2241	0.1681	0.1009

Theorem

Limit $n \rightarrow \infty$ Binomial distribution appears poisson distribution.

(or)

Derive the limiting case of Binomial Distribution as a poisson Distribution.

proof:

We know that

Binomial Distribution

$$B(n, p) = {}^n C_x p^x q^{n-x} \quad (q = 1-p)$$

$$= {}^n C_x p^x (1-p)^{n-x} \quad [p^x = p \cdot p \cdot p \cdot p \cdot p \dots p]$$
$$= \frac{n(n-1)(n-2) \dots (n-(x-1))}{x!} p^x (1-p)^{n-x}$$

$$= \frac{np(np-p)(np-2p) \dots (np-(x-1)p)}{x!} (1-p)^{n-x}$$

We know that $\boxed{\lambda = np} \Rightarrow \left[\frac{\lambda}{n} = p \right]$

$$= \frac{\lambda \left(\lambda - \frac{\lambda}{n} \right) \left(\lambda - \frac{2\lambda}{n} \right) \dots \left(\lambda - \frac{(x-1)\lambda}{n} \right)}{x!} \left(1 - \frac{\lambda}{n} \right)^{n-x}$$

$$= \frac{1 \left(1 - \frac{1}{n} \right) \left(1 - \frac{2}{n} \right) \dots \left(1 - \frac{(x-1)}{n} \right) \lambda^x}{x!} \frac{\left(1 - \frac{\lambda}{n} \right)^n}{\left(1 - \frac{\lambda}{n} \right)^x}$$

Taking $n \rightarrow \infty$

$$= 1 \cdot \frac{\lambda^x}{x!} \frac{(1 - \lambda/n)^n}{(1 - \lambda/n)^x}$$

as $n \rightarrow \infty$ and $p = 0$

$$= \frac{\lambda^x}{x!} \frac{e^{-\lambda}}{1}$$

$$B(n, p) = \frac{e^{-\lambda} \lambda^x}{x!} = p(x = \lambda)$$

$$\left[\begin{array}{l} \because \lim_{n \rightarrow \infty} (1 - \frac{\lambda}{n})^n = 1 \\ \because \lim_{n \rightarrow \infty} (1 - \frac{\lambda}{n})^n = e^{-\lambda} \end{array} \right]$$

$$\lim_{n \rightarrow \infty} (1 - \frac{\lambda}{n})^n = (1 - 0)^n = 1$$

problem B.D

out of 800 families with 5 children each how many would you expect to have either 2 or 3 boys?

Sol Given $n = 5$

Given $N = 800$ families

Let $X \sim B(n, p)$

$$B(n, p) = P(X = x) = n C x p^x q^{n-x} \quad x = 0, 1, 2, 3, \dots$$

$S = \{ \text{Boys, Girls} \}$

$$n(S) = 2$$

$$P(\text{Boys}) = 1/2 \text{ (or)} = 1 - p = 1/2$$

$$(i) p(x=3) = nC_x p^x q^{n-x} = {}^5C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^{5-3}$$

$$p(x=3) = 5/16$$

$$(ii) p(x \leq 3) = p(x=2) + p(x=3)$$

$$\text{No of families} = 800 \times \frac{5}{8} = 500 \text{ families}$$

Normal Distribution

Definition: A Random variable x is said to have a Normal Distribution if its density is given by $f(x, \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-1/2 \left(\frac{x-\mu}{\sigma}\right)^2}$ $\rightarrow \text{---}$

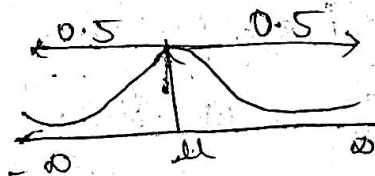
Where μ is the mean and σ is the standard deviation of x .

Note: (i) The Random variable x is then said to be a normal random variable (or) normal variate.

The curve representation the normal distribution (i) is called the normal curve and the total area bounded by the curve and x -axis is one.

i.e.,

$$\int_{-\infty}^{\infty} f(x) dx = 1$$



(2) mean = mode = median = μ .

Hence the distribution is symmetrical

properties of Normal Distribution $X \sim N(\mu, \sigma)$

1. The graph of the Normal distribution $y = f(x)$ in the xy -plane is known as the normal curve

2) The curve is a bell shaped curve and symmetric with respect to mean.

3) The Area under the normal curve represents the total population

4) mean, median, and mode of the distribution coincide at $x = \mu$ as the distribution is symmetric

5) x -axis is an asymptote to the curve

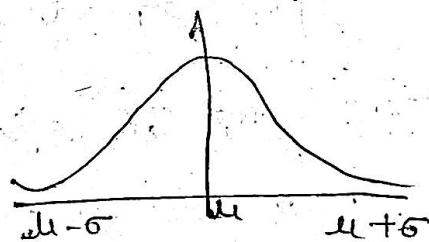
Standard Normal Distribution

$X \sim N(\mu, \sigma) \rightarrow$ Normal Distribution

$X \sim N(0, 1) \rightarrow$ Standard Normal Distribution

Uses of Normal Distribution

1. The Normal Distribution to approximate Binomial and Poisson distribution



2. It has extensive use in sampling theory

To find probability density of Normal curve

The probability that the normal variate x with mean μ & standard deviation σ lies between two specific values x_1 and x_2 with $x_1 < x_2$ can be obtained using area under the standard normal curve as follows

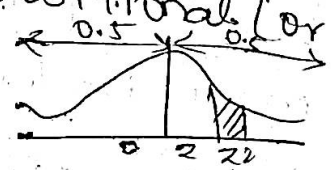
Step 1 is perform the change the of scale $z = \frac{x - \mu}{\sigma}$ and find z_1 & z_2 corresponding to the

values x_1 & x_2 respectively.

50

Step-2: To find $P(x_1 \leq X \leq x_2) = P(z_1 \leq Z \leq z_2)$

Case-1: If both z_1 & z_2 are positional (or both negative) then



$$P(x_1 \leq X \leq x_2) = P(0 \leq Z \leq z_2) - P(0 \leq Z \leq z_1)$$

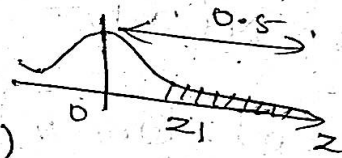
Case-2: If $z_1 < 0$ & $z_2 > 0$ then



$$P(x_1 \leq X \leq x_2) = P(0 \leq Z \leq z_2) + P(0 \leq Z \leq z_1)$$

Step-2(b): To find $P(Z > z_1)$

Case-i: If $z_1 \geq 0$ then



$$P(Z > z_1) = 0.5 - P(0 \leq Z \leq z_1)$$

Case-ii: If $z_1 < 0$ then

$$P(Z > z_1) = 0.5 + P(0 \leq Z \leq z_1)$$



Step-2(c): To find $P(Z \leq z_1) = 1 - P(Z > z_1)$

$$P(Z \leq z_1) = 1 - P(Z > z_1) = 1 - (0.5 - P(0 \leq Z \leq z_1))$$

$$= 0.5 + P(0 \leq Z \leq z_1)$$

Case-2: If $z \leq 0$ then



$$P(Z < z_1) = 1 - P(Z > z_1) = 1 - (0.5 + P(0 \leq Z \leq z_1))$$

$$= 0.5 - P(0 \leq Z \leq z_1)$$

∴ If X is a Normal variate with mean 30 & standard deviation 5, Find

(i) $P(26 \leq X \leq 40)$

(ii) $P(X \geq 45)$

50) let $x \sim N(\mu, \sigma)$

Given mean $= \mu = 30$

standard deviation $= \sigma = 5$

To find: $P(26 \leq x \leq 40)$

We know that, $z = \frac{x - \mu}{\sigma}$

$$\text{When } x = 26, z = \frac{26 - 30}{5} = -\frac{4}{5} = -0.8$$

$$x = 40, z = \frac{40 - 30}{5} = \frac{10}{5} = 2$$

Now,

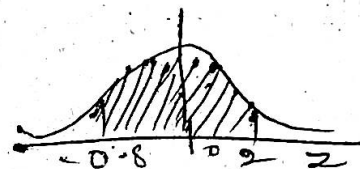
$$P(26 \leq x \leq 40)$$

$$= P(-0.8 \leq z \leq 2)$$

$$= P(-0.8 \leq z \leq 0) + P(0 \leq z \leq 2)$$

$$= P(0 \leq z \leq 0.8) + P(0 \leq z \leq 2)$$

$$= 0.47 + 0.2881 = 0.7653$$



(ii) $P(x \geq 45)$

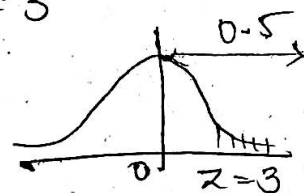
$$z = \frac{x - \mu}{\sigma} = \frac{45 - 30}{5} = \frac{15}{5} = 3$$

Now, $P(x \geq 45)$

$$= 0.5 - P(0 \leq z \leq 3)$$

$$= 0.5 - 0.4987$$

$$= \underline{0.0013}$$



Q. In a Normal Distribution 31% of the items are under 45 and 8% are over 64. Find the mean and variance of the distribution

Sol Given

$$P(X < 45) = 31\% = \frac{31}{100} = 0.31$$

$$P(X > 64) = 8\% = \frac{8}{100} = 0.08$$

We know that

$$Z = \frac{X - \mu}{\sigma}$$

When $X = 45$, $Z_1 = \frac{45 - \mu}{\sigma}$

$$\frac{45 - \mu}{\sigma} = -Z_1 \text{ (Left of the ordinate)}$$

$$\frac{45 - \mu}{\sigma} = -0.5 \text{ [see table [0.19]]}$$

$$45 - \mu = 0.5\sigma \rightarrow (1)$$

When $X = 64$, $Z_2 = \frac{64 - \mu}{\sigma}$

$$\frac{64 - \mu}{\sigma} = Z_2 \text{ [Right of the ordinate]}$$

$$\frac{64 - \mu}{\sigma} = 1.4 \text{ [see table (0.42)]}$$

$$64 - \mu = 1.4\sigma \rightarrow (2)$$

Solve (2) - (1)

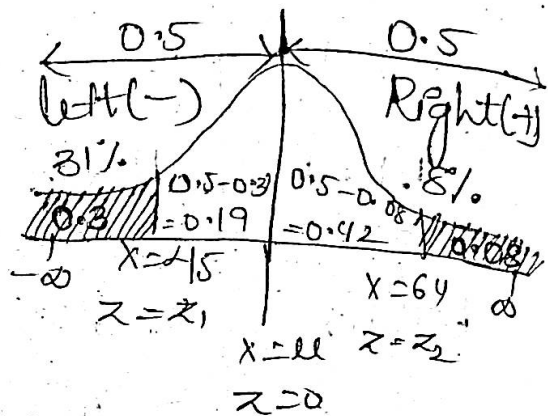
$$64 - \mu = 1.4\sigma$$

$$45 - \mu = -0.5\sigma$$

$$\begin{array}{r} 64 - \mu = 1.4\sigma \\ -(45 - \mu = -0.5\sigma) \\ \hline 19 = 1.9\sigma \end{array}$$

$$\sigma = \frac{19}{1.9} = 10$$

$$\boxed{\sigma = 10}$$



put $\sigma = 10$, in ①

$$45 - \mu = -0.5 \sigma = -0.5(10)$$

$$45 - \mu = -5$$

$$-\mu = -5 - 45$$

$$-\mu = -50$$

$$\boxed{\mu = 50}$$

$$\text{mean} = 50$$

$$\text{standard deviation} = 10$$

8. In a Normal Distribution 7% of the items are under 35 and 89% are under 63. Find the mean and Variance of the distribution

Sol Given

$$p(x < 35) = 7\% = \frac{7}{100} = 0.07 \quad [0.5 - 0.07]$$

$$p(x < 63) = 89\% = \frac{89}{100} = 0.89 \quad [0.5 - 0.89]$$

We know that $p(x > a) = 1 - p(x < a)$

$$\text{Now } P(x < 63) = 1 - p(x < 63) = 1 - 0.89 = 0.11$$

$$\text{WKT when } z = \frac{x - \mu}{\sigma}$$

$$\text{When } x = 35, \quad z = \frac{35 - \mu}{\sigma}$$

$$\frac{35 - \mu}{\sigma} = -z \quad (\text{left of the ordinate})$$

$$\frac{35 - \mu}{\sigma} = -1.5 \quad [\text{see table } (0.45)]$$
$$\frac{35 - \mu}{10} = -1.5 \quad [0.5 - 0.07]$$

$$35 - \mu = -15 \rightarrow \text{①}$$

$$\text{When } x = 63, \quad z = \frac{63 - \mu}{\sigma}$$

$$\frac{63 - \mu}{\sigma} = z_2 \quad (\text{Right of the ordinate})$$

$$\frac{63 - \mu}{\sigma} = 1.2 \quad (\text{see table } (0.39) (0.5 - 0.89))$$

$$63 - \mu = 1.26 \rightarrow (2)$$

Solve (2) - (1)

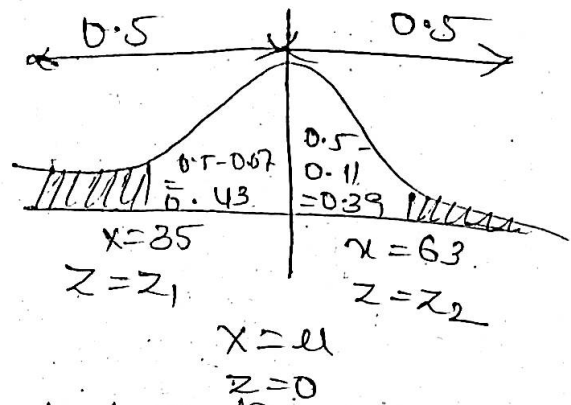
$$63 - \mu = 1.26$$

$$\begin{array}{r} 35 - \mu = -1.56 \\ (-) \quad (+) \quad (+) \\ \hline \end{array}$$

$$28 = 2.76$$

$$\sigma = \frac{28}{2.7} = 10.3$$

$\sigma = 10$ Standard deviation



put $\sigma = 10$ in (2)

$$63 - \mu = 1.26$$

$\mu = 50.3$ mean

3) 1000 students have written an examination. The mean of test is 35 and standard deviation is 5. Assuming the distribution to be normal. Find

(i) $P(25 \leq x \leq 40)$ (ii) $P(x \geq 40)$

(iii) $P(x \leq 20)$ (iv) $P(x \geq 50)$

4) Given that mean height of students in a class is 158 centimeters and standard deviation of 20 cm. Find how many students height lies between 150 cm and 170 cm if there are 100 students in the class $\boxed{0.3811}$

5) If the mean mass of 300 students are normally distribution with mean 68 kgs and standard deviation 3 kgs. How many students have mass.

(i) greater than 72 kg (ii) lesser than equal to 64 kg (iii) between 65 and 71 kg .